

fog Fluctuations of Glaciers (FoG) Database

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Internationally collected, standardized dataset on glacier state and change (length, area, elevation, mass) based on in-situ measurements, remote sensing, and reconstructions. This file is generated from contents of the `datapackage.json` file.

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- **publications** WGMS (2023): Global Glacier Change Bulletin No. 5 (2020-2021). Michael Zemp, Isabelle Gärtner-Roer, Samuel U. Nussbaumer, Ethan Z. Welty, Inés Dussailant, and Jacqueline Bannwart (eds.), ISC (WDS) / IUGG (IACS) / UNEP / UNESCO / WMO, World Glacier Monitoring Service, Zurich, Switzerland, 134 pp. Based on database version <https://doi.org/10.5904/wgms-fog-2023-09>.

WGMS (2013): Glacier Mass Balance Bulletin No. 12 (2010-2011). Michael Zemp, Samuel U. Nussbaumer, Kathrin Naegeli, Isabelle Gärtner-Roer, Frank Paul, Martin Hoelzle, and Wilfried Haeberli (eds.), ICSU (WDS) / IUGG (IACS) / UNEP / UNESCO / WMO, World Glacier Monitoring Service, Zurich, Switzerland, 106 pp. Based on database version <https://doi.org/10.5904/wgms-fog-2013-11>.

WGMS (2012): Fluctuations of Glaciers 2005-2010 (Vol. X): Michael Zemp, Holger Frey, Isabelle Gärtner-Roer, Samuel U. Nussbaumer, Martin Hoelzle, Frank Paul, and Wilfried Haeberli (eds.), ICSU (WDS) / IUGG (IACS) / UNEP / UNESCO / WMO, World Glacier Monitoring Service, Zurich, Switzerland. Based on database version <https://doi.org/10.5904/wgms-fog-2012-11>.

... and earlier issues (https://wgms.ch/literature_published_by_wgms)
- **contributors** WGMS scientific collaboration network of national correspondents and principal investigators as listed in the data (`investigators` and `agencies` columns) and related publications.
- **disclaimer** The data may contain errors and inaccuracies. Hence, we strongly suggest performing data quality checks and, in case of ambiguities, to contact us as well as the investigators and agencies listed in the data (`investigators` and `agencies` columns) and related publications.

1 Overview

The data tables are stored as comma-separated value (CSV) files in the `data` subfolder. Tables refer to each other by special columns that uniquely identify each record (i.e., a foreign key referencing another table's primary key).

- `glacier`: Glacier lookup by country, region, names, latitude and longitude, and `GLIMS`, `RGI`, and `WGI` identifiers. The `id` column uniquely identifies a glacier and is referenced in other tables using their `glacier_id` column.
- `state`: Glacier-wide area, elevation (mean, lowest, highest), length, and terminus type. For area and mean elevation by elevation band, see `state_band` (whose `state_id` references column `id` in `state`).
- `change`: Glacier-wide elevation and volume change (typically from repeat mapping of surface elevation). For elevation and volume change by elevation band, see `change_band` (whose `change_id` references column `id` in `change`).
- `front_variation`: Changes in glacier length (typically from repeat measurements to a fixed reference).
- `mass_balance`: Glacier-wide mass balance and survey overview. See `mass_balance_band` for mass balance by elevation band and `mass_balance_point` for the original point measurements. Both refer to the `mass_balance` table by columns `glacier_id` and `year`.
- `event`: Extraordinary events concerning glaciers.
- `person`: Persons referenced from other tables' `investigators` column.
- `agency`: Agencies referenced from other tables' `investigators` and `agencies` columns.

2 Major changes

2.1 2025-02b

- Fix column `mass_balance_point.longitude`, whose values were all equal to `latitude`.

2.2 2025-02

- **All measurement units are standardized to the meter:** m, m w.e., m², m³, kg m⁻³.
- Accumulation-area ratio (`mass_balance.aar`) is a fraction (0-1) rather than a percentage (0-100%).
- All table and column names are all lowercase rather than all uppercase.
- The glacier identifier `WGMS_ID` is renamed `glacier_id` in the data tables and references column `id` in table `glacier`. For clarity, it should be referred to externally as the `wgms_fog_glacier_id` or equivalent.
- All dates are formatted as `yyyy-mm-dd` (no more `99` for unknown month and/or day) and accompanied by an uncertainty column in days. For example, `date: 1986-09-30` and `date_unc: 1` represents the interval 1986-09-29 to 1986-10-01.
- Tables `mass_balance` (glacier-wide and elevation-band balance) and `mass_balance_overview` (overview) are renamed `mass_balance` (overview and glacier-wide balance) and `mass_balance_band` (elevation-band balance).
- New table `state_band` stores area and mean elevation by elevation band.
- Table `change` is split into tables `change` (glacier-wide) and `change_band` (elevation-band). Area changes are moved to timeseries of area in the `state` and `state_band` tables.
- Glacier outlines are associated to glacier-wide measurements with column `outline_id` or columns `begin_outline_id` and `end_outline_id`. These columns report either a `GLIMS` analysis id (e.g. `glims:754359`), an `RGI` id (`RGI2000-v7.0-G-11-02773`), or the integer id of a custom outline (stored in the `outline` table, available upon request).
- Columns `investigators` and `agencies` use a consistent format and reference records in the new tables `person` and `agency`.

3 Tables

glacier · state · state_band · change · change_band · front_variation · mass_balance · mass_balance_band · mass_balance_point · event · person · agency

3.1 glacier

General information about each glacier.

- path: data/glacier.csv
- primaryKey: id
- foreignKeys
 - parent_glacier_id → glacier.id

name	type	description
country <i>required</i>	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the glacier is (primarily) located.
short_name <i>required</i>	string	The short name only contains characters A-Z, 0-9, - (dash), . (period), : (colon), () (parentheses), / (forward slash), ' (apostrophe), and (space). Non-Latin characters are transliterated, and if the Latin name contains accents, the following rules are typically applied: Å → AA, Æ → AE, Ä → AE, ð → D, Ø → OE, œ → OE, Ö → OE, ß → SS, þ → TH, Ü → UE (neglecting any remaining accents).
names	string	Full names (in English and/or local languages) as a pipe-delimited list.
id <i>required</i>	integer	Integer identifying the glacier in the Fluctuations of Glaciers (FoG) database.
latitude <i>required</i>	number (degrees)	Latitude (EPSG:4326). Positive values indicate the northern hemisphere and negative values indicate the southern hemisphere. The point (latitude , longitude) should be in the main channel in the upper part of the glacier ablation area.
longitude <i>required</i>	number (degrees)	Longitude (EPSG:4326). Positive values indicate east of the zero meridian and negative values indicate west of the zero meridian. The point (latitude , longitude) should be in the main channel in the upper part of the glacier ablation area.
gtng_region	string	Global Terrestrial Network for Glaciers (GTN-G) 2023 code (https://10.5904/gtng-glacreg-2023-07) of the region in which the glacier is located.
glims_id	string	Glacier identifier (glac_id) in the Global Land Ice Measurements from Space database (https://www.glims.org). It has the format GxxxxxxEyyyyyθ , where xxxxxx is longitude east of the zero meridian in millidegrees, yyyy is north or south latitude in millidegrees, and θ is N or S depending on the hemisphere.

name	type	description
rgi50_ids	string	Glacier identifiers (RGIId), as a pipe-delimited list, from the Randolph Glacier Inventory 5.0 (https://nsidc.org/data/nsidc-0770/versions/5). These have format RGI50-rr.nnnnn , where rr is the first-order region (zero-padded) and nnnnn is an arbitrary numeric code.
rgi60_ids	string	Glacier identifiers (RGIId), as a pipe-delimited list, from the Randolph Glacier Inventory 6.0 (https://nsidc.org/data/nsidc-0770/versions/6). These have format RGI60-rr.nnnnn , where rr is the first-order region (zero-padded) and nnnnn is an arbitrary numeric code.
rgi70_ids	string	Glacier or glacier complex identifiers (rgi_id), as a pipe-delimited list, from the Randolph Glacier Inventory 7.0 (https://nsidc.org/data/nsidc-0770/versions/7). These have format RGI2000-v7.0-p-rr-nnnnn , where p is the product code (G: glacier or C: complex), rr is the first-order region (zero-padded), and nnnnn is an arbitrary numeric code.
wgi_id	string	Glacier identifier (wgi_glacier_id) in the World Glacier Inventory (https://nsidc.org/data/g01130/versions/1). It is constructed from the following elements: - 2-character political unit - 1-character continent code - 4-character drainage code - 2-character free position code - 3-character local glacier code
parent_glacier_id	integer	Identifier (id) of the (present or former) parent glacier.
references	string	Literature references as a pipe-delimited list.
remarks	string	Remarks and flags as a pipe-delimited list.

3.2 state

Measurements of glacier state – length, area, elevation, and terminus type.

- path: data/state.csv
- primaryKey: id
- foreignKeys
 - country · glacier_name · glacier_id → glacier.country · glacier.short_name · glacier.id

name	type	description
country <i>required</i>	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the glacier is (primarily) located.
glacier_name <i>required</i>	string	Short name of the glacier (see glacier.short_name).
glacier_id <i>required</i>	integer	Integer identifying the glacier in the Fluctuations of Glaciers (FoG) database (see glacier.id).
outline_id	string	Glacier outline used to derive the reported measurement(s), as either a GLIMS (https://www.glims.org) analysis id prefixed with 'glims:', RGI (https://www.glims.org/RGI) id, or integer id of a custom outline (available upon request).
id <i>required</i>	integer	Integer identifying the glacier state.
date <i>required</i>	date (yyyy-mm-dd)	Date of the reported measurements.
date_unc	number (days)	Estimated (or exact) range of date (e.g. 2020-10-02 ± 1 days = 2020-10-01 to 2020-10-03).
highest_elevation	integer (m)	Highest elevation on the glacier.
lowest_elevation	integer (m)	Lowest elevation on the glacier.
mean_elevation	integer (m)	Mean glacier elevation.
elevation_unc	integer (m)	Estimated random error of reported elevations.
area	integer (m ²)	Glacier area.
area_unc	integer (m ²)	Estimated random error of area .
length	integer (m)	Glacier length measured along the main flowline.

name	type	description
length_unc	integer (m)	Estimated random error of length .
terminus_type	string	Condition at the terminus. - land: Dry land - cliff: Very steep dry land (subject to dry calving) - lake: Lake (freshwater) - marine: Sea or ocean (saltwater) - shelf: Ice shelf (which itself terminates in the ocean)
platform	string	Survey platform. - ground: Terrestrial - air: Airborne - space: Spaceborne - other: Other (see remarks)
method	string	Survey method. - direct: In-situ measurement (e.g. GPS, tape measure) - map: Maps and mapping - photo: Photographs and photogrammetry - laser: Laser scanning - radar: Radar - reconstruction: Reconstruction (from historical or dating methods) - historical: Historical (from texts or visuals) - text: Text - print: Print (from an engraving or etching). For photographic prints, use photo. - painting: Painting - drawing: Drawing - radiocarbon: Radiocarbon date - dendrochronology: Dendrochronology - surface exposure: Surface exposure dating (e.g. TCN) - other: Other (see remarks)
investigators	string	Pipe-delimited list of persons (linked to person table as person.name) and their agency affiliations (linked to agencies and agency table as agency.id) responsible for the measurements.
agencies	string	Numbered (by agency.id), pipe-delimited list of agencies referred to from investigators .
references	string	Literature references as a pipe-delimited list.
remarks	string	Remarks and flags as a pipe-delimited list.

3.3 state_band

Measurements of glacier state by elevation band.

- path: data/state_band.csv
- primaryKey: state_id · lower_elevation · upper_elevation
- foreignKeys
 - glacier_id · state_id → state.glacier_id · state.id
 - country · glacier_name · glacier_id → glacier.country · glacier.short_name · glacier.id

name	type	description
country <i>required</i>	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the glacier is (primarily) located.
glacier_name <i>required</i>	string	Short name of the glacier (see <code>glacier.short_name</code>).
glacier_id <i>required</i>	integer	Integer identifying the glacier in the Fluctuations of Glaciers (FoG) database (see <code>glacier.id</code>).
state_id <i>required</i>	integer	Integer identifying the glacier state.
lower_elevation <i>required</i>	integer (m)	Lower boundary of the surface elevation band.
upper_elevation <i>required</i>	integer (m)	Upper boundary of the surface elevation band.
mean_elevation	integer (m)	Mean glacier elevation.
elevation_unc	integer (m)	Estimated random error of reported elevations.
area	integer (m ²)	Elevation band area.
area_unc	integer (m ²)	Estimated random error of <code>area</code> .
remarks	string	Remarks and flags as a pipe-delimited list.

3.4 change

Measurements of glacier elevation change – typically from geodetic surveys.

- path: data/change.csv
- primaryKey: id
- foreignKeys
 - country · glacier_name · glacier_id → glacier.country · glacier.short_name · glacier.id

name	type	description
country <i>required</i>	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the glacier is (primarily) located.
glacier_name <i>required</i>	string	Short name of the glacier (see glacier.short_name).
glacier_id <i>required</i>	integer	Integer identifying the glacier in the Fluctuations of Glaciers (FoG) database (see glacier.id).
begin_outline_id	string	Glacier outline used to derive the glacier state on begin_date , as either a GLIMS (https://www.glims.org) analysis id prefixed with 'glims:', RGI (https://www.glims.org/RGI) id, or integer id of a custom outline (available upon request).
end_outline_id	string	Glacier outline used to derive the glacier state on end_date , as either a GLIMS (https://www.glims.org) analysis id prefixed with 'glims:', RGI (https://www.glims.org/RGI) id, or integer id of a custom outline (available upon request).
id <i>required</i>	integer	Integer identifying the glacier change.
begin_date <i>required</i>	date (yyyy-mm-dd)	Date of the earlier (reference) measurement.
begin_date_unc	number (days)	Estimated (or exact) range of begin_date (e.g. 2020-10-02 ± 1 days = 2020-10-01 to 2020-10-03).
end_date <i>required</i>	date (yyyy-mm-dd)	Date of the later measurement.
end_date_unc	number (days)	Estimated (or exact) range of end_date (e.g. 2021-10-02 ± 1 days = 2021-10-01 to 2021-10-03).
area	integer (m ²)	Glacier area (presumably from begin_date , end_date , or an average) which should be used to estimate elevation_change from volume_change or vice versa.
elevation_change	number (m)	Change in surface elevation (averaged over the glacier) from begin_date to end_date .
elevation_change_unc	number (m)	Estimated random error of elevation_change .

name	type	description
volume_change	integer (m ³)	Change in glacier volume from <code>begin_date</code> to <code>end_date</code> .
volume_change_unc	integer (m ³)	Estimated random error of <code>volume_change</code> .
begin_platform	string	Survey platform used on <code>begin_date</code> . - ground: Terrestrial - air: Airborne - space: Spaceborne - other: Other (see <code>remarks</code>)
begin_method	string	Survey method used on <code>begin_date</code> . - direct: In-situ measurement (e.g. GPS, tape measure) - map: Maps and mapping - photo: Photographs and photogrammetry - laser: Laser scanning - radar: Radar - reconstruction: Reconstruction (from historical or dating methods) - historical: Historical (from texts or visuals) - text: Text - print: Print (from an engraving or etching). For photographic prints, use photo. - painting: Painting - drawing: Drawing - radiocarbon: Radiocarbon date - dendrochronology: Dendrochronology - surface exposure: Surface exposure dating (e.g. TCN) - other: Other (see <code>remarks</code>)
end_platform	string	Survey platform used on <code>end_date</code> . - ground: Terrestrial - air: Airborne - space: Spaceborne - other: Other (see <code>remarks</code>)

name	type	description
end_method	string	Survey method used on end_date . - direct: In-situ measurement (e.g. GPS, tape measure) - map: Maps and mapping - photo: Photographs and photogrammetry - laser: Laser scanning - radar: Radar - reconstruction: Reconstruction (from historical or dating methods) - historical: Historical (from texts or visuals) - text: Text - print: Print (from an engraving or etching). For photographic prints, use photo. - painting: Painting - drawing: Drawing - radiocarbon: Radiocarbon date - dendrochronology: Dendrochronology - surface exposure: Surface exposure dating (e.g. TCN) - other: Other (see remarks)
investigators	string	Pipe-delimited list of persons (linked to person table as person.name) and their agency affiliations (linked to agencies and agency table as agency.id) responsible for the measurements.
agencies	string	Numbered (by agency.id), pipe-delimited list of agencies referred to from investigators .
references	string	Literature references as a pipe-delimited list.
remarks	string	Remarks and flags as a pipe-delimited list.

3.5 change_band

Measurements of glacier elevation change – typically from geodetic surveys – by elevation band.

- path: data/change_band.csv
- primaryKey: change_id · lower_elevation · upper_elevation
- foreignKeys
 - glacier_id · change_id → change.glacier_id · change.id
 - country · glacier_name · glacier_id → glacier.country · glacier.short_name · glacier.id

name	type	description
country <i>required</i>	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the glacier is (primarily) located.
glacier_name <i>required</i>	string	Short name of the glacier (see <code>glacier.short_name</code>).
glacier_id <i>required</i>	integer	Integer identifying the glacier in the Fluctuations of Glaciers (FoG) database (see <code>glacier.id</code>).
change_id <i>required</i>	integer	Integer identifying the glacier change.
lower_elevation <i>required</i>	integer (m)	Lower boundary of the surface elevation band.
upper_elevation <i>required</i>	integer (m)	Upper boundary of the surface elevation band.
area	integer (m ²)	Elevation band area (presumably from <code>begin_date</code> , <code>end_date</code> , or an average) which should be used to estimate <code>elevation_change</code> from <code>volume_change</code> or vice versa.
elevation_change	number (m)	Change in surface elevation (averaged over the elevation band) from <code>begin_date</code> to <code>end_date</code> .
elevation_change_unc	number (m)	Estimated random error of <code>elevation_change</code> .
volume_change	integer (m ³)	Change in glacier volume (in the elevation band) from <code>begin_date</code> to <code>end_date</code> .
volume_change_unc	integer (m ³)	Estimated random error of <code>volume_change</code> .
remarks	string	Remarks and flags as a pipe-delimited list.

3.6 front_variation

Measurements of glacier front variation (length change).

- path: data/front_variation.csv
- primaryKey: glacier_id · series_id · end_date
- foreignKeys
 - country · glacier_name · glacier_id → glacier.country · glacier.short_name · glacier.id

name	type	description
country <i>required</i>	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the glacier is (primarily) located.
glacier_name <i>required</i>	string	Short name of the glacier (see glacier.short_name).
glacier_id <i>required</i>	integer	Integer identifying the glacier in the Fluctuations of Glaciers (FoG) database (see glacier.id).
series_id <i>required</i>	integer	Integer used to distinguish between different front variation series for the same glacier. Value 0 is reserved for the longterm in-situ series (if one exists).
begin_date	date (yyyy-mm-dd)	Date of the earlier (reference) measurement, or null if no reference was used (e.g. glacier advance inferred from moraine condition).
begin_date_unc	number (days)	Estimated (or exact) range of begin_date (e.g. 2020-10-02 ± 1 days = 2020-10-01 to 2020-10-03).
end_date <i>required</i>	date (yyyy-mm-dd)	Date of the later measurement.
end_date_unc	number (days)	Estimated (or exact) range of end_date (e.g. 2021-10-02 ± 1 days = 2021-10-01 to 2021-10-03).
length_change	number (m)	Change in glacier length from begin_date to end_date (+ advance, - retreat).
length_change_unc	number (m)	Estimated random error of length_change .
length_change_direction	string	Length change direction (in the absence of a quantitative length_change) from begin_date to end_date . - advance: Glacier advanced (+) - retreat: Glacier retreated (-) - stationary: Glacier was stationary (0)
end_platform	string	Survey platform used on end_date . - ground: Terrestrial - air: Airborne - space: Spaceborne - other: Other (see remarks)

name	type	description
end_method	string	Survey method used on <code>end_date</code> . - direct: In-situ measurement (e.g. GPS, tape measure) - map: Maps and mapping - photo: Photographs and photogrammetry - laser: Laser scanning - radar: Radar - reconstruction: Reconstruction (from historical or dating methods) - historical: Historical (from texts or visuals) - text: Text - print: Print (from an engraving or etching). For photographic prints, use photo. - painting: Painting - drawing: Drawing - radiocarbon: Radiocarbon date - dendrochronology: Dendrochronology - surface exposure: Surface exposure dating (e.g. TCN) - other: Other (see <code>remarks</code>)
investigators	string	Pipe-delimited list of persons (linked to <code>person</code> table as <code>person.name</code>) and their agency affiliations (linked to <code>agencies</code> and <code>agency</code> table as <code>agency.id</code>) responsible for the measurements.
agencies	string	Numbered (by <code>agency.id</code>), pipe-delimited list of agencies referred to from <code>investigators</code> .
references	string	Literature references as a pipe-delimited list.
remarks	string	Remarks and flags as a pipe-delimited list.

3.7 mass_balance

Glacier-wide mass balance and survey details.

- path: data/mass_balance.csv
- primaryKey: glacier_id · year
- foreignKeys
 - country · glacier_name · glacier_id → glacier.country · glacier.short_name · glacier.id

name	type	description
country <i>required</i>	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the glacier is (primarily) located.
glacier_name <i>required</i>	string	Short name of the glacier (see glacier.short_name).
glacier_id <i>required</i>	integer	Integer identifying the glacier in the Fluctuations of Glaciers (FoG) database (see glacier.id).
outline_id	string	Glacier outline used to derive the reported glacier-wide mass balance(s), as either a GLIMS (https://www.glims.org) analysis id prefixed with 'glims:', RGI (https://www.glims.org/RGI) id, or integer id of a custom outline (available upon request).
year <i>required</i>	year	Calendar year associated with the last accumulation (winter) - ablation (summer) cycle. This is almost always the calendar year at the end of the measurement period unless the cycle extends only briefly into the following year (e.g. 2020-01-05 to 2021-01-17 ends in 2021 but is the 2020 hydrological year).
time_system	string	Time measurement system of the dates (begin_date, midseason_date and end_date) and balances (winter_balance, summer_balance, annual_balance). See Cogley 2011 (https://unesdoc.unesco.org/ark:/48223/pf0000192525) for background. - floating: Floating-date. Dates are the dates of measurement, and balances are as measured. [Introduced in 2011, so earlier balances based on this system are theoretically reported as 'other'. Redefined in 2024 to include stratigraphic measurements.] - fixed: Fixed-date. Dates are those to which the measured balances were corrected (typically the hydrological year), and balances are as corrected. Correction methods should be given in the remarks. [Redefined to include stratigraphic measurements in 2024] - stratigraphic: Stratigraphic. Involves measurements between successive minimum or maximum surfaces (at least in the accumulation area), often in combination with fixed or floating-date measurements in the ablation area. [Discontinued in 2024] - combined: Some combination of the above (see remarks). [Discontinued in 2024] - other: Other (see remarks).

name	type	description
begin_date	date (yyyy-mm-dd)	Begin (reference) date of the reported <code>winter_balance</code> and <code>annual_balance</code> .
begin_date_unc	number (days)	Estimated (or exact) range of <code>begin_date</code> (e.g. 2020-10-02 ± 1 days = 2020-10-01 to 2020-10-03).
midseason_date	date (yyyy-mm-dd)	End date of the reported <code>winter_balance</code> and begin date of the reported <code>summer_balance</code> .
midseason_date_unc	number (days)	Estimated (or exact) range of <code>midseason_date</code> (e.g. 2021-05-04 ± 2 days = 2021-05-02 to 2021-05-06).
end_date	date (yyyy-mm-dd)	End date of the reported <code>summer_balance</code> and <code>annual_balance</code> .
end_date_unc	number (days)	Estimated (or exact) range of <code>end_date</code> (e.g. 2021-10-02 ± 1 days = 2021-10-01 to 2021-10-03).
winter_balance	number (m w.e.)	Mass balance over the winter (accumulation) season – from <code>begin_date</code> to <code>midseason_date</code> .
winter_balance_unc	number (m w.e.)	Estimated random error of <code>winter_balance</code> .
summer_balance	number (m w.e.)	Mass balance over the summer (ablation) season – from <code>midseason_date</code> to <code>end_date</code> .
summer_balance_unc	number (m w.e.)	Estimated random error of <code>summer_balance</code> .
annual_balance	number (m w.e.)	Mass balance over the hydrological year – from <code>begin_date</code> to <code>end_date</code> .
annual_balance_unc	number (m w.e.)	Estimated random error of <code>annual_balance</code> .
ela_position	string	Position of the equilibrium line altitude (ELA) relative to the glacier. - on: On the glacier - above: Above the glacier (<code>ela</code> may be a lower bound) - below: Below the glacier (<code>ela</code> may be an upper bound)
ela	integer (m)	Mean elevation, averaged over the glacier, of the end-of-mass-balance-year equilibrium line. This can be outside the elevation range of the glacier if the <code>ela_position</code> was <code>below</code> or <code>above</code> .
ela_unc	integer (m)	Estimated random error of <code>ela</code> .
aar	number	Accumulation area divided by the total glacier area.
area	integer (m ²)	Area considered in the calculation of the glacier-wide mass balance.

name	type	description
investigators	string	Pipe-delimited list of persons (linked to <code>person</code> table as <code>person.name</code>) and their agency affiliations (linked to <code>agencies</code> and <code>agency</code> table as <code>agency.id</code>) responsible for the measurements.
agencies	string	Numbered (by <code>agency.id</code>), pipe-delimited list of agencies referred to from <code>investigators</code> .
references	string	Literature references as a pipe-delimited list.
remarks	string	Remarks and flags as a pipe-delimited list.

3.8 mass_balance_band

Glacier mass balance by elevation band.

- path: data/mass_balance_band.csv
- primaryKey: glacier_id · year · lower_elevation · upper_elevation
- foreignKeys
 - country · glacier_name · glacier_id → glacier.country · glacier.short_name · glacier.id
 - glacier_id · year → mass_balance.glacier_id · mass_balance.year

name	type	description
country <i>required</i>	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the glacier is (primarily) located.
glacier_name <i>required</i>	string	Short name of the glacier (see <code>glacier.short_name</code>).
glacier_id <i>required</i>	integer	Integer identifying the glacier in the Fluctuations of Glaciers (FoG) database (see <code>glacier.id</code>).
year <i>required</i>	year	Calendar year associated with the last accumulation (winter) - ablation (summer) cycle. This is almost always the calendar year at the end of the measurement period unless the cycle extends only briefly into the following year (e.g. 2020-01-05 to 2021-01-17 ends in 2021 but is the 2020 hydrological year).
lower_elevation <i>required</i>	integer (m)	Lower boundary of the surface elevation band.
upper_elevation <i>required</i>	integer (m)	Upper boundary of the surface elevation band.
area	integer (m ²)	Area of the elevation band.
winter_balance	number (m w.e.)	Mass balance over the winter (accumulation) season – from <code>mass_balance.begin_date</code> to <code>mass_balance.midseason_date</code> .
winter_balance_unc	number (m w.e.)	Estimated random error of <code>winter_balance</code> .
summer_balance	number (m w.e.)	Mass balance over the summer (ablation) season – from <code>mass_balance.midseason_date</code> to <code>mass_balance.end_date</code> .
summer_balance_unc	number (m w.e.)	Estimated random error of <code>summer_balance</code> .
annual_balance	number (m w.e.)	Mass balance over the hydrological year – from <code>mass_balance.begin_date</code> to <code>mass_balance.end_date</code> .
annual_balance_unc	number (m w.e.)	Estimated random error of <code>annual_balance</code> .
remarks	string	Remarks and flags as a pipe-delimited list.

3.9 mass_balance_point

Glacier mass balance measured at specific points (e.g. stakes or pits).

- path: data/mass_balance_point.csv
- primaryKey: id
- foreignKeys
 - country · glacier_name · glacier_id → glacier.country · glacier.short_name · glacier.id
 - glacier_id · year → mass_balance.glacier_id · mass_balance.year

name	type	description
country <i>required</i>	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the glacier is (primarily) located.
glacier_name <i>required</i>	string	Short name of the glacier (see <code>glacier.short_name</code>).
glacier_id <i>required</i>	integer	Integer identifying the glacier in the Fluctuations of Glaciers (FoG) database (see <code>glacier.id</code>).
year <i>required</i>	year	Calendar year associated with the last accumulation (winter) - ablation (summer) cycle. This is almost always the calendar year at the end of the measurement period unless the cycle extends only briefly into the following year (e.g. 2020-01-05 to 2021-01-17 ends in 2021 but is the 2020 hydrological year).
id <i>required</i>	integer	Unique identifier for the point.
original_id	string	Identifier for the point used by the original investigators.
begin_date <i>required</i>	date (yyyy-mm-dd)	Begin (reference) date of the reported <code>balance</code> .
begin_date_unc	number (days)	Estimated range of <code>begin_date</code> (e.g. 2020-10-02 ± 1 days = 2020-10-01 to 2020-10-03).
end_date <i>required</i>	date (yyyy-mm-dd)	End date of the reported <code>balance</code> .
end_date_unc	number (days)	Estimated range of <code>end_date</code> (e.g. 2021-10-02 ± 1 days = 2021-10-01 to 2021-10-03).
latitude	number (degrees)	Point latitude (EPSG:4326). Positive values indicate the northern hemisphere and negative values indicate the southern hemisphere.
longitude	number (degrees)	Point longitude (EPSG:4326). Positive values indicate east of the zero meridian and negative values indicate west of the zero meridian.
elevation	integer (m)	Glacier surface elevation at point.
balance	number (m w.e.)	Point mass balance from <code>begin_date</code> to <code>end_date</code> (+ accumulation, - ablation).

name	type	description
balance_unc	number (m w.e.)	Estimated random error of balance .
density	integer (kg m ⁻³)	Measured or estimated density used to convert height change to mass balance. If multiple density values were used (e.g. for snow and ice), they should be described in remarks .
density_unc	integer (kg m ⁻³)	Estimated random error of density .
balance_code required	string	Whether and how the point balance was used in the calculation of glacier-wide (mass_balance) and elevation band (mass_balance_band) balance. - annual: Used for annual balance. - summer: Used for summer balance. - winter: Used for winter balance. - index: Not used (index measurement).
remarks	string	Remarks and flags as a pipe-delimited list.

3.10 event

Noteworthy events concerning glacier hazards and dramatic glacier changes.

- path: data/event.csv
- primaryKey: id
- foreignKeys
 - country · glacier_name · glacier_id → glacier.country · glacier.short_name · glacier.id

name	type	description
country <i>required</i>	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the glacier is (primarily) located.
glacier_name	string	Short name of the glacier (see glacier.short_name).
glacier_id <i>required</i>	integer	Integer identifying the glacier in the Fluctuations of Glaciers (FoG) database (see glacier.id).
id <i>required</i>	integer	Integer indentifying the event.
date	string	Date of the event. Events that span multiple days or represent a series of events should be further described with dates in description .
date_unc	number (days)	Estimated (or exact) range of date (e.g. 2020-10-02 ± 1 days = 2020-10-01 to 2020-10-03).
latitude	number (degrees)	Latitude (EPSG:4326). Positive values indicate the northern hemisphere and negative values indicate the southern hemisphere. The point (latitude , longitude) should be as close as possible to the event source.
longitude	number (degrees)	Longitude (EPSG:4326). Positive values indicate east of the zero meridian and negative values indicate west of the zero meridian. The point (latitude , longitude) should be as close as possible to the event source.
description	string	Summary description of the event sequence - including for example the type and scale of the damage, measures taken to mitigate glacier hazards, and studies carried out in connection with the event. Quantitative information should be included whenever possible. - Surge: Date and location of onset, duration, flow velocity, discharge anomalies and periodicity - Calving: Rate of retreat, iceberg discharge, flow velocity and water depth at calving front - Flood: Volume, mechanism, peak discharge, sediment load, reach and propagation velocity of flood wave or flow front - Ice avalanche: Volume, runout distance, overall slope (ratio of vertical drop height to horizontal runout distance) of path - Tectonics: Volumes, runout distances and overall slopes (ratio of vertical drop height to horizontal runout distance) of rockfall on glacier surface, amount of geothermal melting in craters, etc.

name	type	description
surge	boolean	Whether a surge was involved.
calving	boolean	Whether calving was involved.
flood	boolean	Whether a flood (e.g. glacial-lake outburst flood) was involved.
avalanche	boolean	Whether an ice avalanche was involved.
rockfall	boolean	Whether rockfall was involved.
debris_flow	boolean	Whether a debris flow was involved.
earthquake	boolean	Whether an earthquake was involved.
volcanic_eruption	boolean	Whether a volcanic eruption was involved.
other	boolean	Whether any other event types were involved.
investigators	string	Pipe-delimited list of persons (linked to <code>person</code> table as <code>person.name</code>) and their agency affiliations (linked to <code>agencies</code> and <code>agency</code> table as <code>agency.id</code>) responsible for the measurements.
agencies	string	Numbered (by <code>agency.id</code>), pipe-delimited list of agencies referred to from <code>investigators</code> .
references	string	Literature references as a pipe-delimited list.
remarks	string	Remarks and flags as a pipe-delimited list.

3.11 person

Details about persons associated with measurements in the database.

- path: data/person.csv
- primaryKey: id

name	type	description
id <i>required</i>	integer	Integer identifying the person.
name <i>required</i>	string	Full name in Latin script.
surname <i>required</i>	string	Surname or family name in Latin script.
original_name	string	Full name in non-Latin script.
synonyms	string	Pipe-delimited list of alternative names in any script.
orcid	string	Open Researcher and Contributor ID (ORCID, convertible to a url as <code>https://orcid.org/{orcid}</code>).
url	string	Profile URL.
birth_year	year	Year of birth.
death_year	year	Year of death.
remarks	string	Remarks and flags as a pipe-delimited list.

3.12 agency

Details about agencies associated with measurements in the database.

- path: data/agency.csv
- primaryKey: id
- foreignKeys
 - parent_agency_id → agency.id
 - former_agency_id → agency.id

name	type	description
country	string	ISO 3166 alpha-2 code (https://www.iso.org/obp/ui) of the country in which the agency is (primarily) located.
id <i>required</i>	integer	Integer identifying the agency.
parent_agency_id	integer	Integer (id) of the (present or former) parent agency.
name <i>required</i>	string	Name.
abbreviation	string	Abbreviation of name .
alternate_name	string	Alternate name (for regions with two official languages).
alternate_abbreviation	string	Abbreviation of alternate_name .
english_name	string	English name (if name or alternate_name is not in English).
english_abbreviation	string	Abbreviation of english_name .
url	string	Homepage URL.
wikipedia_url	string	Wikipedia URL (English if available).
wikidata_id	string	Wikidata identifier (convertable to a url as https://wikidata.org/wiki/{wikidata_id}).
begin_year	year	Year of founding or the renaming of former_agency_id .
end_year	year	Year of dissolution or renaming.
former_agency_id	integer	Integer (id) of the agency that this agency was formerly known as.
remarks	string	Remarks and flags as a pipe-delimited list.